

Second-Order Bragg Scattering Theory for Time-Evolving Nonlinear Sea Surfaces

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Abstract—A facet-based second-order Bragg scattering theory for nonlinear sea surface scattering is proposed in this letter. The discretized sea surface is a composition of facets with appropriate size, which is large enough in comparison with the radar wavelength and sufficiently small as it can still be regarded as a plane. These small facets are described as a superposition of a group of cosine waves modulated by capillary waves with Bragg and second-order Bragg resonance wavelengths. The contribution of Bragg scattering, second-order Bragg scattering, and their interactions are considered using integral term transformation. The backscattering coefficients from the 2-D Creamer II sea surface in the C- and X-bands are comprehensively analyzed and compared between radar measurements and numerical predictions, demonstrating the capabilities of the theory. The Doppler spectrum results indicate that both the nonlinear nature of the sea surface and the second-order Bragg scattering effects could significantly broaden the Doppler spectrum and shift the Doppler centroid.

Index Terms—Doppler spectrum, nonlinear sea surfaces, scattering, second-order Bragg scattering theory.

I. INTRODUCTION

THE electromagnetic-wave scattering from sea surfaces has been of particular interest in ocean remote sensing, target detection, and identification in the marine environment for over half a century, both in theoretical and experimental terms. With the advancement of radar technology, there are more and more active remote sensing observations in microwave bands [1], [2]. In particular, the Doppler scatterometer shows great potential for exploring ocean surface motion retrievals such as ocean vector wind and ocean vector motion [3].

Although some progress has been made, the simulation of the scattering coefficient and Doppler spectrum from the sea surface is still indispensable to better understand the physical mechanism of sea scattering characteristics due to its low cost and easy realization [4], [5]. A full polarimetric sea-surface scattering model combining the SSA2 theory with the

vector radiative transfer theory was presented in [6]. However, this scattering model is statistical, which is applicable for wind retrieval, but not for Doppler analysis and ocean surface motion retrieval. X-band is widely used in both tactical and remote-sensing radar systems [7]. However, for Doppler simulations using the traditional numerical method, the facet dimension should be no more than $\lambda/10$. For example, for a wavelength of 3 cm, the number of sample points would be $33\,333 \times 33\,333$ in the sea area of $100 \times 100 \text{ m}^2$. Thus, the traditional numerical method may not be suitable to take charge of the time-evolving realizations with computationally expensive Doppler simulations in X-band. Wang and Xu [8] applied the weighted curvature approximation method to simulate the scattering fields of the 2-D time-evolving sea surfaces by decomposing the Fourier spectrum of 2-D sea surface into multiple blocks, which provided an excellent attempt to calculate the Doppler spectrum in X-band. An applicable and feasible facet model based on the Bass and Fuks' two-scale model [9] with capillary wave modification was proposed, with a facet dimension of $1 \times 1 \text{ m}^2$, which can calculate the Doppler spectra of time-evolving sea surfaces in the C-band without the need for such a large number of sample points [10].

On the other hand, for medium incident angle, the sea surface scattering echo is mainly considered as Bragg scattering, which is applicable to the simulation results of the sea surface scattering coefficient in the majority of cases. However, the experimental data show that there will be Doppler spectrum broadening and frequency shift at C- and X-band compared with L-band, which has not been clearly explained by the physical mechanism [11]. One of the reasons for this phenomenon is the influence of the nonlinear hydrodynamic effects increasing with radar frequency [12]. The high-order Bragg scattering effects are investigated in this letter and may be another important reason for the significant broadening of the Doppler spectrum and a shift of the Doppler centroid.

This work is organized as follows. Section II gives a brief introduction to Creamer II nonlinear sea surface generation based on the Elfouhaily spectrum and Hilbert transformation. The discretized sea surface profile with small facets described as a superposition of a group of cosine waves modulated by capillary wave with Bragg and second-order Bragg resonance wavelength is also established in this section. In Section III, a derivation of the second-order Bragg scattering theory for the discretized nonlinear sea surface is given. In Section IV, we present the results of numerical simulations and comparisons of scattering characteristics with the experimental data. The Doppler spectra are also presented in this section. Finally, in Section V the conclusions are drawn.

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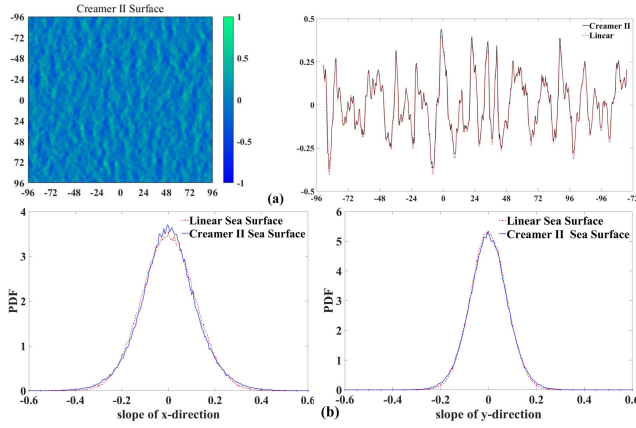


Fig. 1. Simulated sea surfaces at time $t = 0$ s. (a) Creamer II surface and comparison of the 1-D profile along x-axis with linear surface. (b) PDF of the slope of the height of the sea surface between linear waves and Creamer II waves.

II. NONLINEAR SEA SURFACE GENERATION

A. Nonlinear Sea Surface

Linear sea surface can reflect the geometry of waves including gravity waves and tension waves to a certain extent, but it is quite different from the real dynamic wave which is especially manifested on the Doppler spectrum induced by the nonlinear hydrodynamic interactions between waves. Therefore, nonlinear models such as the Creamer model [13] and choppy wave model (CWM) [14] that can better reflect the weak interaction between waves have been proposed and applied to sea surface electromagnetic scattering. The Creamer II model is actually more strongly nonlinear than CWM inasmuch as it exhibits stronger slopes and curvatures and produces enhanced Doppler shifts. Thus, we use the Creamer II model to generate the nonlinear sea surfaces in this letter. The Creamer formulation writes as a nonlinear transformation of the Hilbert transform of the linear surface as follows:

$$\zeta(\mathbf{r}, t) = \text{Re} \sum_{\mathbf{k}} \left(-i \frac{\mathbf{k}}{k} \right) F_L(\mathbf{k}; t) \exp(i\mathbf{k} \cdot \mathbf{r}) \quad (1)$$

where $F_L(\mathbf{k}; t)$ is the complex amplitude of Fourier components of the linear surface. After expanding the exponential operator as a series, the complex amplitude of Fourier components of nonlinear sea surface can be expressed as

$$A_{\text{NL}}(\mathbf{k}, t) = \frac{1}{N} \sum_{\mathbf{r}} \frac{\exp[i\mathbf{k} \cdot \zeta(\mathbf{r}, t)] - 1}{k} \exp(-i\mathbf{k} \cdot \mathbf{r}). \quad (2)$$

The first order of this series identifies is with $F_L(\mathbf{k}; t)$, while the second order is given by

$$A_{\text{NL}}^{(2)}(K_x, K_y; t) = -\frac{K_x^2}{2K} F^D(\zeta_{0x}^2) - \frac{K_x K_y}{K} F^D(\zeta_{0x} \zeta_{0y}) - \frac{K_y^2}{2K} F^D(\zeta_{0y}^2) \quad (3)$$

where $F^D(\cdot)$ refers to 2-D Fourier transform. Therefore, the Creamer II surface can be obtained by IFFT of $F_L(k_x, k_y; t) + A_{\text{NL}}^{(2)}(k_x, k_y; t)$.

Fig. 1 shows the simulated sea surfaces at time $t = 0$ s. The wind-fetch length of the Elfouhaily spectrum is 50 km, and the wind speed and direction are 5 m/s and 180° , respectively. The area of the surface is 192×192 m and the discrete sampling points along the x- and y-directions are $M = N = 256$.

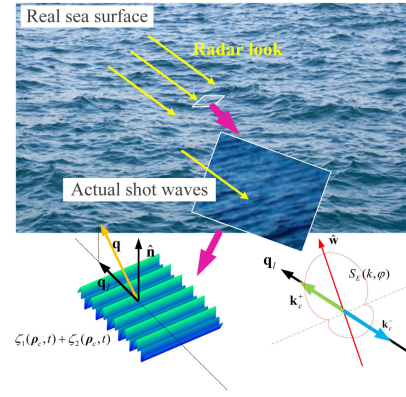


Fig. 2. Geometry of the faceted two-scale profile: long waves and capillary wave tilted by long waves.

It can be seen from the 1-D profile that with the consideration of the nonlinear interaction between waves, the wave crest of the nonlinear sea surface is steeper, the wave trough is smoother, and the nonlinear sea surface has more small-scale ripples. The slope distributions in the x- and y-directions for the Creamer II sea surface and the linear surface sea surface are presented in Fig. 1. We observe that there are more facets with large slope values for the Creamer II sea surface than that for linear surface, which has a significant effect on the scattering characteristics and Doppler spectrum of the surface. It is noted that a spectrum undressing procedure is typically necessary before surface generation [14], which is neglected due to its slight effect on the scattering results compared with the nonlinear hydrodynamic effects and second-order Bragg scattering emphasized in this work.

B. Small Scale Wave Modeling

As shown in Fig. 2, when the radar irradiates the real ocean surface, it mainly detects those capillary wave components along its observation direction which satisfies Bragg resonance conditions. We simplify this group of components into the superposition of a group of cosine waves with resonance wavelength and second-order Bragg resonance wavelength, that is,

$$\begin{aligned} \zeta(\rho_c, t) &= \zeta_1(\rho_c, t) + \zeta_2(\rho_c, t) \\ &= B_1(\mathbf{k}_{c1}) \cos(\mathbf{k}_{c1} \cdot \rho_c - \omega_{c1} t) \\ &\quad + B_2(\mathbf{k}_{c2}) \cos(\mathbf{k}_{c2} \cdot \rho_c - \omega_{c2} t) \end{aligned} \quad (4)$$

where $\mathbf{k}_{c1}$ and $\mathbf{k}_{c2} = 2\mathbf{k}_{c1}$ is the Bragg and second-order Bragg resonant wave wavenumber vector, respectively. $\mathbf{k}_c$ is defined along the local scattering vector $\mathbf{q}_l$, which is the projection of $\mathbf{q} = \mathbf{k}(\hat{\mathbf{k}}_s - \hat{\mathbf{k}}_i)$ on the tilt plane. $\mathbf{k}_c$ contains the sum of wavenumber components $\mathbf{k}_c^+$ and $\mathbf{k}_c^-$ relative to the positive and negative directions. Considering that the capillary wave components far from and close to the radar will contribute to radar echo, it is not difficult to understand that the wavenumber vector $\mathbf{k}_c$ will contain the wavenumber components in both positive and negative directions relative to $\mathbf{q}_l$. Thus $\mathbf{k}_c^\pm = (k_x^\pm, k_y^\pm)$ can be expressed as

$$\begin{aligned} k_{cx}^\pm &= \pm 2\pi q_l q_{ly} / (\lambda_c |q_l|^2) \\ k_{cy}^\pm &= \pm 2\pi q_l q_{lx} / (\lambda_c |q_l|^2). \end{aligned} \quad (5)$$

In (4), $B_1(\mathbf{k}_{c1})$ and $B_2(\mathbf{k}_{c2})$ is the amplitude of capillary wave, which can be defined as the corresponding energy amplitude

in its wave spectrum related to Bragg and second-order Bragg resonance wavenumber

$$B_i(\mathbf{k}_{ci}) = 2\pi\sqrt{S_E^{\text{capi}}(\mathbf{k}_{ci})/\Delta x\Delta y}, \quad i = 1, 2 \quad (6)$$

where $S_E^{\text{capi}}(\mathbf{k}_c)$ is the capillary spectrum located in the higher part of the full Elfouhaily 2-D sea spectrum

$$S_E^{\text{capi}}(\mathbf{k}) = \begin{cases} 0, & |\mathbf{k}| < \mathbf{k}_{\text{cut}} \\ S_E(\mathbf{k}), & |\mathbf{k}| \geq \mathbf{k}_{\text{cut}}. \end{cases} \quad (7)$$

$\mathbf{k}_{\text{cut}}$ is the truncated wavenumber. We choose $\mathbf{k}_{\text{cut}} = k/20$ here, $\Delta x\Delta y$ is the area of the small facet. ω_c in (4) is the circular frequency corresponding to the wavenumber k_c and t is the time factor. $\boldsymbol{\rho}_c = (x_c, y_c)$ is the coordinate at the profile within a facet and the height superimposed on the small facet can be expressed as

$$z^{\text{capi}} = f(\boldsymbol{\rho}_0, t) + \zeta_1(\boldsymbol{\rho}_c, t) + \zeta_2(\boldsymbol{\rho}_c, t) + Z_x x_c + Z_y y_c. \quad (8)$$

$f(\boldsymbol{\rho}_0, t)$ is the height of the center of the large-scale facet, $\boldsymbol{\rho}_0 = (x_g, y_g)$ is the position vector of the small facet center. Z_x and Z_y represent the surface slopes in the x - and y -directions.

III. SECOND-ORDER BRAGG SCATTERING THEORY

According to the perturbation theory [10], the scattering field of an arbitrarily inclined small facet can be expressed as

$$E_{PQ}^{\text{facet}}(\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_s) = [\hat{\mathbf{H}}_s, \hat{\mathbf{V}}_s] 2\pi \frac{e^{ikR_0}}{iR_0} \mathbb{S}_{PQ}(\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_s) \quad (9)$$

where $\hat{\mathbf{k}}_i$ is a unit vector directed from the transmitter, while $\hat{\mathbf{k}}_s$ is a unit vector directed to the reception point, $[\hat{\mathbf{H}}_s, \hat{\mathbf{V}}_s]$ is the unit polarization vector. $\mathbb{S}_{PQ}$ is the scattering amplitude matrix, which can be expressed as

$$\mathbb{S}_{PQ}(\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_s) = \frac{k^2(1-\varepsilon)}{8\pi^2} \tilde{\mathbb{F}}_{PQ} \iint \{\zeta_1(\mathbf{r}) + \zeta_2(\mathbf{r})\} e^{-i\mathbf{q} \cdot \mathbf{r}} d\boldsymbol{\rho} \quad (10)$$

where $\tilde{\mathbb{F}}_{PQ}$ is the scattering factor matrix. In (9), the integral phase term modulated by capillary wave including the first- and second-order Bragg wave can be expressed as in (11), shown at the bottom of the page, where $\hat{\mathbf{n}} = (n_x, n_y, n_z)$ is the unit normal vector of the small facet and

$$\begin{aligned} I_{01}(\mathbf{k}_c) &= \sin c \left\{ \frac{\Delta x_g}{2} [(1+m+2n)k_{cx} - q_x - q_z Z_x] \right\} \\ &\quad \cdot \sin c \left\{ \frac{\Delta y_g}{2} [(1+m+2n)k_{cy} - q_y - q_z Z_y] \right\} \\ &\quad \times e^{-i(m+1)\omega_{c1}t} e^{-in\omega_{c2}t} \\ I_{02}(\mathbf{k}_c) &= \sin c \left\{ \frac{\Delta x_g}{2} [(1-m-2n)k_{cx} + q_x + q_z Z_x] \right\} \end{aligned}$$

$$\cdot \sin c \left\{ \frac{\Delta y_g}{2} [(1-m-2n)k_{cy} + q_y + q_z Z_y] \right\} \\ \times e^{i(1-m)\omega_{c1}t} e^{-in\omega_{c2}t}$$

$$\begin{aligned} I_{03}(\mathbf{k}_c) &= \sin c \left\{ \frac{\Delta x_g}{2} [(2+m+2n)k_{cx} - q_x - q_z Z_x] \right\} \\ &\quad \cdot \sin c \left\{ \frac{\Delta y_g}{2} [(2+m+2n)k_{cy} - q_y - q_z Z_y] \right\} \\ &\quad \times e^{-i(n+1)\omega_{c2}t} e^{-im\omega_{c1}t} \\ I_{04}(\mathbf{k}_c) &= \sin c \left\{ \frac{\Delta x_g}{2} [(2-m-2n)k_{cx} + q_x + q_z Z_x] \right\} \\ &\quad \cdot \sin c \left\{ \frac{\Delta y_g}{2} [(2-m-2n)k_{cy} + q_y + q_z Z_y] \right\} \\ &\quad \times e^{i(1-n)\omega_{c2}t} e^{-im\omega_{c1}t}. \end{aligned} \quad (12)$$

Ignoring the multiple scattering of each small facet on the sea surface, the scattering field of the whole sea surface could be expressed in the summation

$$E^s(\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_s, t) = \sum_{m=1}^M \sum_{n=1}^N E_{mn}^{\text{facet}}(\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_s). \quad (13)$$

IV. RESULT AND DISCUSSIONS

A. Backscattering Coefficient

Fig. 3 shows the results and comparisons of first-order and first- and second-order Bragg backscattering coefficient from the Creamer II nonlinear sea surface and the measured data [15], [16] under different polarizations (HH-pol represents that both the incident and scattered waves are horizontally polarized, and VV-pol represents both the incident and scattered waves are vertically polarized.) in the C-band (4.445 GHz) and X-band (8.91 GHz). The simulated parameter for the sea surface is the same with Fig. 1, and the backscattering coefficient was averaged over 50 surface samples. It is noted that the proper selection of the facet dimension is required to ensure the decorrelation of the phasor returns from different facets [17]. Experimentally, the average NRCS does not depend much upon the facet size that changed from 0.3 to 1.5 m. Thus, we reasonably choose a facet size of 0.75 m for both C- and X-band and the sample points are $M = N = 256$ along the x - and y -directions in this work.

We can see that the effect of high-order Bragg scattering reduces the backscattering coefficient at a small incident angle (less than 15°) and increased by about 3–4 dB at medium and large incident angles (large than 30°). Compared with the measured data, we can observe that for all incident angle, especially for medium incident angle, the second-order Bragg scattering theory proposed in this letter is more accurate compared with traditional Bragg scattering. For large incident angle, the non-Bragg scattering effects such as breaking waves

$$\begin{aligned} I(\cdot) &= \int_{\Delta S} \{\zeta_1(\mathbf{r}, t) + \zeta_2(\mathbf{r}, t)\} \exp(-i\mathbf{q} \cdot \mathbf{r}) d\boldsymbol{\rho} = \frac{e^{-i\mathbf{q} \cdot \mathbf{r}_0} \Delta S}{2n_z} \\ &\quad \cdot \left\{ \begin{aligned} &B_1(\mathbf{k}_{c1}) \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} (-i)^{m+n} J_m[q_z B_1(\mathbf{k}_{c1})] J_n[q_z B_2(\mathbf{k}_{c2})] [I_{01}(\mathbf{k}_c) + I_{02}(\mathbf{k})] \\ &+ B_2(\mathbf{k}_{c2}) \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} (-i)^{m+n} J_m[q_z B_1(\mathbf{k}_{c1})] J_n[q_z B_2(\mathbf{k}_{c2})] [I_{03}(\mathbf{k}_c) + I_{04}(\mathbf{k})] \end{aligned} \right\} \end{aligned} \quad (11)$$

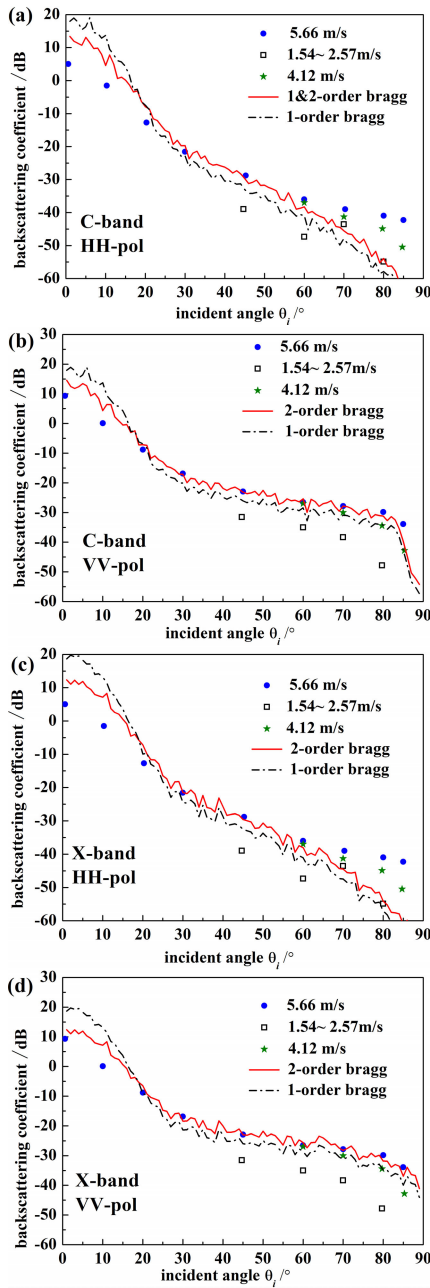


Fig. 3. Backscattering coefficient of Creamer II sea surface with a wind speed of 5 m/s and wind direction of 180° and measured data. (a) HH-pol, C-band. (b) VV-pol, C-band. (c) HH-pol, X-band. (d) VV-pol, X-band.

and foam, and the multi-interaction between the radar wave and ocean surface were analyzed in our previous work [18], which are temporarily beyond this letter.

B. Doppler Spectrum

Fig. 4 shows the results of the linear and nonlinear sea surface Doppler spectra under different polarizations in the C- and X-band. The simulated parameter for the sea surface is the same with Fig. 3 and the Doppler spectra was averaged over 50 surface samples. The simulated time for Doppler spectra was 2.56 s with a time interval of 5 ms. We can see that the Doppler spectrum of nonlinear sea surface has a significant broadening compared to that of the linear sea surface, which is due to the position shift of the small-scale capillaries caused by the nonlinear waves interaction. Besides, the shift of the Doppler spectrum of nonlinear sea surface in HH polarization

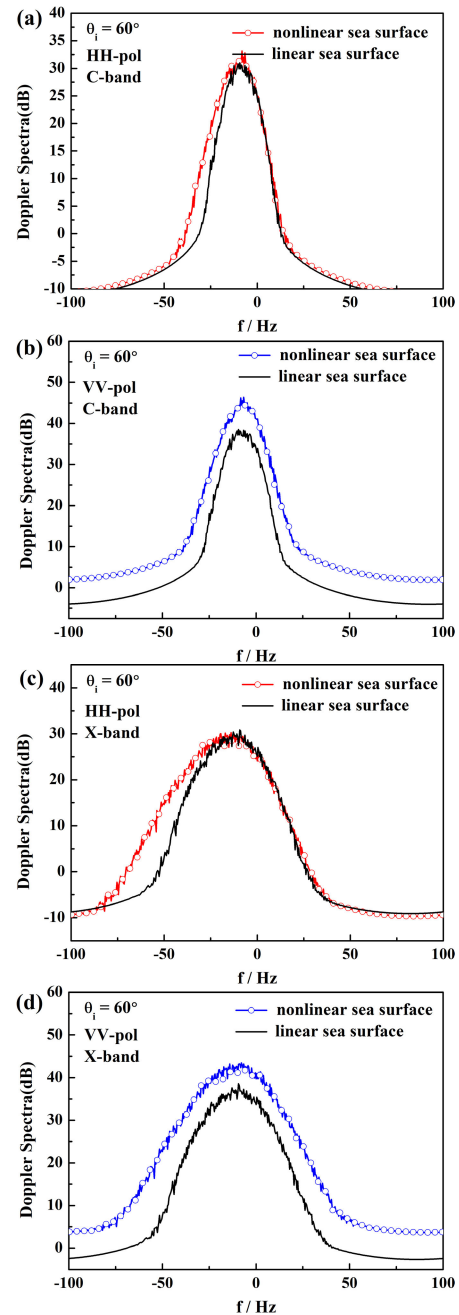


Fig. 4. Comparative Doppler spectra from the linear and nonlinear sea surface. (a) HH-pol, C-band. (b) VV-pol, C-band. (c) HH-pol, X-band. (d) VV-pol, X-band.

is greater than that in VV polarization. By comparing the Doppler spectra in C- and X-band, we can conclude that the influence of the nonlinear hydrodynamic effects increases with radar frequency.

Fig. 5(a) and (b) shows the results of the Doppler spectra with first-order Bragg scattering and first- and second-order Bragg scattering from nonlinear sea surface under different polarizations in C-band. We can see that the Doppler spectrum with second-order Bragg scattering considered shows a significant broadening with two-peaks, which is due to the interaction of the small-scale capillaries composed both by first- and second-order Bragg wave. Fig. 5(c) and (d) shows the results of the Doppler spectra with the first-order Bragg scattering and first- and second-order Bragg scattering from nonlinear sea surface under different polarizations in X-band. The intensity of the second-order peak increases in the X-band,

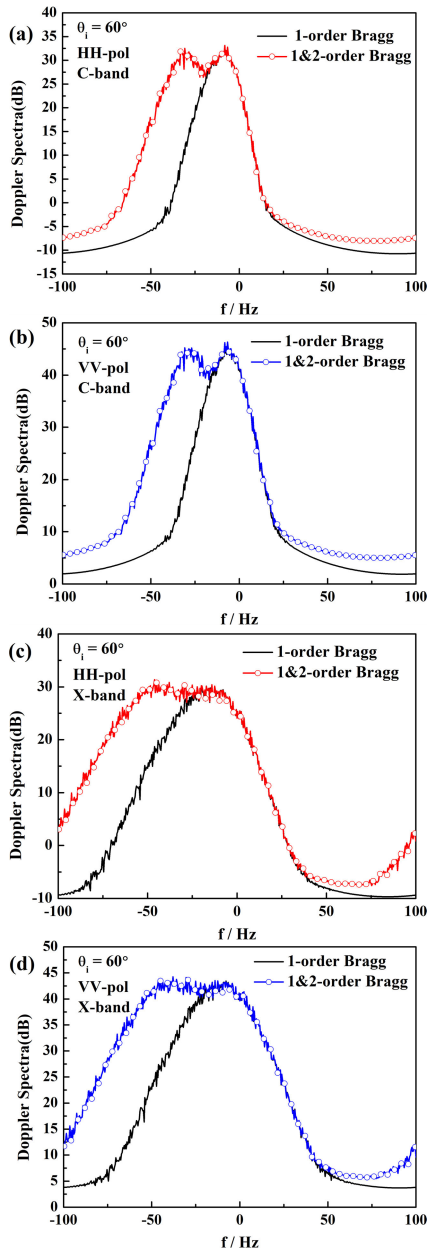


Fig. 5. Comparative Doppler spectra from the first-order and first- and second-order Bragg scattering models. (a) HH-pol, C-band. (b) VV-pol, C-band. (c) HH-pol, X-band. (d) VV-pol, X-band.

which makes the two-peaks of the Doppler spectrum appears to one broadening peak with a more significant shift. We can conclude that the second-order Bragg scattering effect on the Doppler broadening and frequency shift increases with radar frequency.

V. CONCLUSION

This letter focused on the first- and second-order Bragg scattering effect on the scattering and Doppler spectra for the C- and X-band from the nonlinear sea surface. From the numerical result, we can observe that the backscattering coefficient reduced at small incident angles while increased by about 3–4 dB at medium and large incident angles with the consideration of the second-order Bragg scattering effect.

It shows that both the nonlinear wave effect and second-order Bragg scattering effect will cause the broadening and frequency shift of Doppler spectrum. Besides the non-Bragg scattering and the multi-interaction between the radar wave and ocean surface, which remain to be future endeavors to enhance the proposed model, the second-order Bragg scattering theory proposed in this letter could overwhelm the application to electromagnetic scattering solution associated with the electrically large ocean scene, with validity and feasibility, which should help guide radar advancement.

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